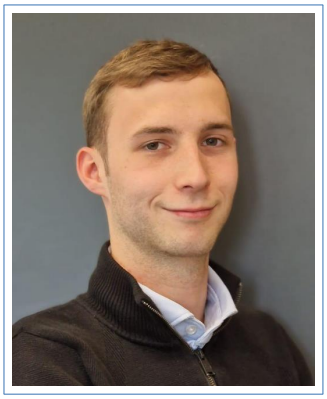
Aaron Kreis

Master's Student in Physics
Expected 2027

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Experimental Physics · Optical Imaging · Scientific Computing

Research Experience

Jun–Aug 2026 **Internship – Automated Bioluminescence Microscopy and Fourier Light-Field Imaging**, *NMSB Group – Prof. Michael Krieg — ICFO – The Institute of Photonic Sciences*

- Expanded and characterized a multimodal microscopy platform for brightfield, fluorescence, bioluminescence, and Fourier light-field microscopy (FLFM)
- Integrated a motorized filter wheel and designed/modified high-precision CAD models for optomechanical improvements to the microscope
- Developed Micro-Manager plugins for automated spiral sample scanning and configurable multi-step acquisition workflows controlling camera, XY/Z stages, illumination, and filter wheel
- Characterized the FLFM system using experimental point-spread-function measurements and reconstructed volumetric C. elegans data using Richardson–Lucy deconvolution; additionally evaluated deep-learning-based LUCID reconstruction models

Jul–Oct 2025 **Bachelor Thesis – Light Ray Propagation in Microscopic Imaging of Red Blood Cells**, *Dynamics of Fluids Group – Prof. Christian Wagner — Saarland University*

- Developed ray tracing simulations to model light propagation in microscopic imaging
- Simulated optical interactions with rotationally symmetric biological structures
- Performed quantitative intensity analysis on the detector plane
- Designed a computational workflow for reproducible microscopy simulations

Scientific Contributions

Journal Articles

2026 L.-F. Morales-Curiel, **A. Kreis**, and M. Krieg. *Seeing in the Dark: Intelligent Fourier Light Field Imaging for Bioluminescence Microscopy. ACS Photonics*, accepted for publication (bioRxiv preprint).

Conference Posters

2026 **A. Kreis**, S. T. Hermes, T. John, and C. Wagner. *Ray-based simulation and experimental analysis of red blood cell imaging in brightfield microscopy*. DPG Spring Meeting, Dresden, Germany, poster BP 14.28.

2025 **A. Kreis**, S. T. Hermes, T. John, and C. Wagner. *Red Blood Cells under brightfield microscopy*. DPG Spring Meeting, Regensburg, Germany, poster BP 17.16.

2025 **A. Kreis** and T. John. *Brownsche Bewegung, optische Pinzetten und numerische Simulationen*. DPG School "Physikalische Praktika" (AGPP), Bad Honnef, Germany (laboratory education poster).

Awards & Grants

2025 – 2027 **Glückauf Scholarship (awarded twice)** — RAG-Stiftung

2026 **Santander Scholarship for Erasmus+ Mobility** — Santander Deutschland

2025 & 2026 **Travel Grants (awarded twice) – DPG Spring Meetings (SKM)** — WE Heraeus-Stiftung

Technical Skills

Programming Python, MATLAB, C++

Engineering CAD, 3D printing

Tools LaTeX, Git, Linux

Education

2025 – 2027 (expected) **M.Sc. Physics (Double Degree)** — Université Grenoble Alpes & Saarland University

2021 – 2025 **B.Sc. Physics** — Saarland University

2013 – 2021 **AbiBac** — Warndt-Gymnasium

Teaching Experience

2023 – 2026 **Course Instructor** — Saarland University

- Summer School "From Atoms to Zettabytes" for STEM-oriented pre-university students
- Mathematics Preparatory Course for incoming undergraduate students

2022 – 2025 **Teaching Assistant for Undergraduate Physics Courses** — Saarland University

- Led problem-solving sessions accompanying undergraduate lectures
- Guided students in experimental methods, data analysis, and scientific reporting
- Supervised laboratory courses and tutorial sessions in mechanics, electromagnetism, and computational physics

Academic Service & Leadership

2022 – present **Student Representative** — Saarland University
Examination Board (B.Sc./M.Sc. Physics); Appointment Committees; Faculty Council; Student Council

2025 – 2026 **Vice President for Partnerships** – PhITEM Student Council — Université Grenoble Alpes

2024 – 2025 **Founding Deputy Chair** – jDPG Regional Group Saarbrücken — German Physical Society

2024 **External Review Panel Member** – B.Sc. Physics Program — University of Luxembourg

2023 – present **Member**, German Physical Society (DPG)

Employment

2022 – 2025 **Student Assistant – Physics Outreach** — Department of Physics – Saarland University

- Designed and delivered physics demonstrations for school visits, career fairs, and public events
- Represented the department in student recruitment initiatives

2021 – 2023 **Student Assistant – Process Automation** — DEKRA Automobil GmbH

- Developed automation workflows to streamline administrative processes across multiple departments
- Reduced manual workload by optimizing repetitive tasks

Languages

German Native

English ≥B2

French ≥B2